

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 4
2 February – 8 February 2026

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Nanyang Technological University
12 January – 8 December 2026



Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



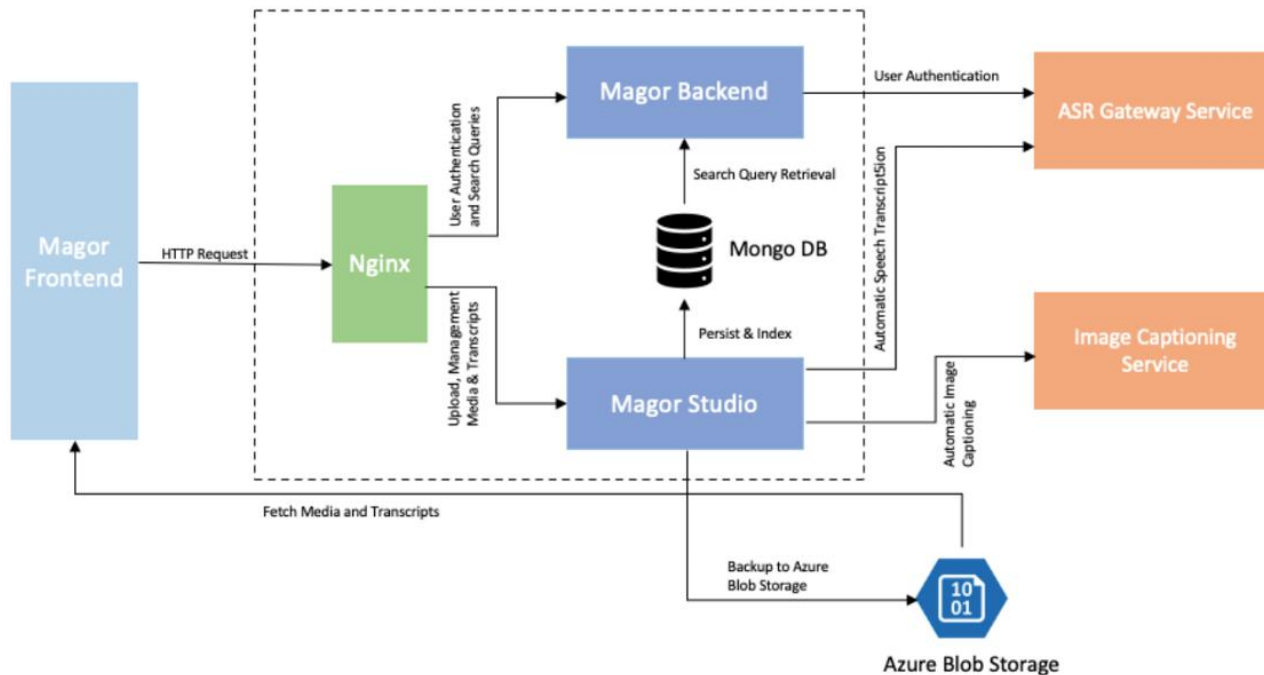
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

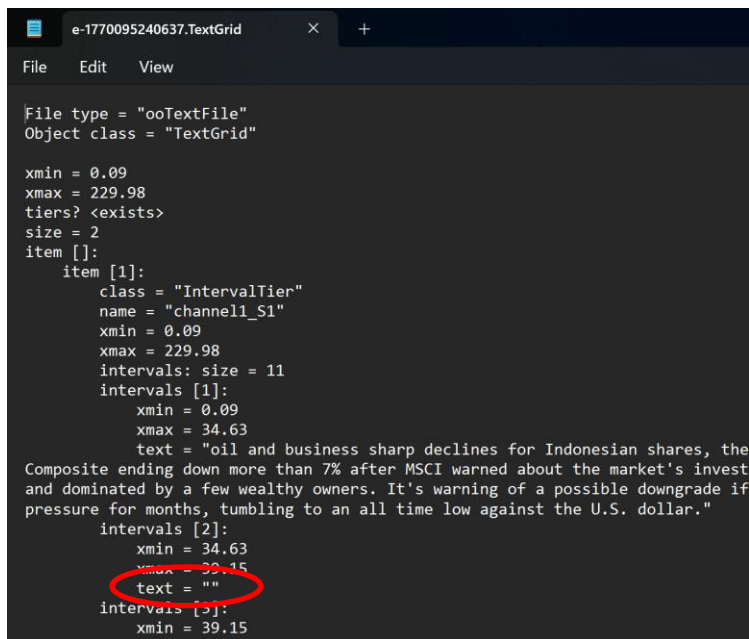
1. Fixed TextGrid and srt conversion format
2. Verified transcript edits reflect in downloads
3. Aligned recording list columns
4. Fixed unintended tag removal on backspace
5. Removed download button when uploading media



Task 1 (Fixed TextGrid and srt conversion format)

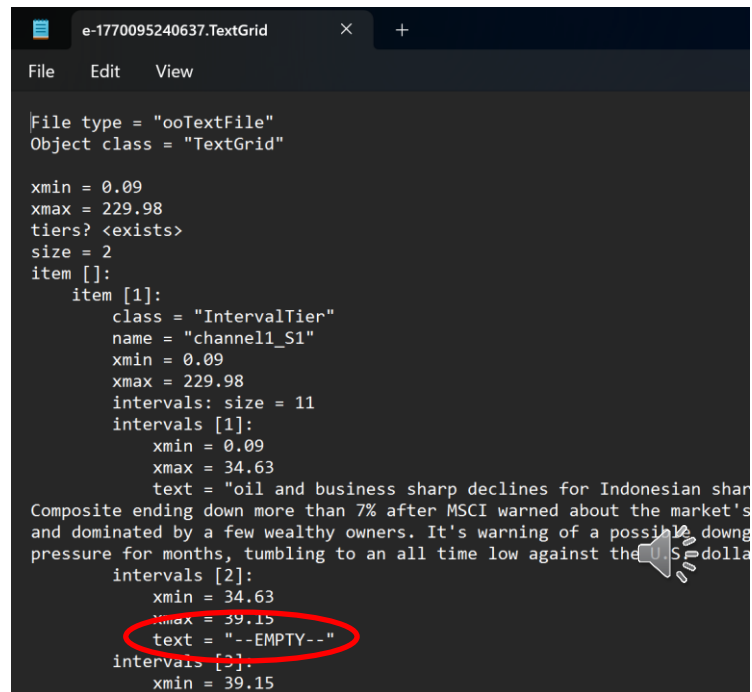
- TextGrid

- Replaced empty strings with “--EMPTY--”



```
File type = "ooTextFile"
Object class = "TextGrid"

xmin = 0.09
xmax = 229.98
tiers? <exists>
size = 2
item []:
  item [1]:
    class = "IntervalTier"
    name = "channell1_S1"
    xmin = 0.09
    xmax = 229.98
    intervals: size = 11
    intervals [1]:
      xmin = 0.09
      xmax = 34.63
      text = "oil and business sharp declines for Indonesian shares, the
Composite ending down more than 7% after MSCI warned about the market's invest
and dominated by a few wealthy owners. It's warning of a possible downgrade if
pressure for months, tumbling to an all time low against the U.S. dollar."
    intervals [2]:
      xmin = 34.63
      xmax = 39.15
      text = ""
    intervals [3]:
      xmin = 39.15
```

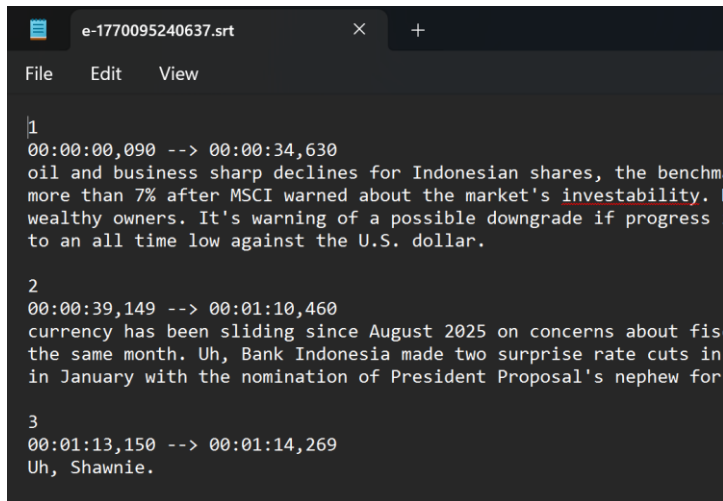


```
File type = "ooTextFile"
Object class = "TextGrid"

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xmax = 229.98
tiers? <exists>
size = 2
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Composite ending down more than 7% after MSCI warned about the market's
and dominated by a few wealthy owners. It's warning of a possible downg
pressure for months, tumbling to an all time low against the U.S. dolla
    intervals [2]:
      xmin = 34.63
      xmax = 39.15
      text = "--EMPTY--"
    intervals [3]:
      xmin = 39.15
```

Task 1 (Fixed TextGrid and srt conversion format)

- srt
 - Added speaker id before text

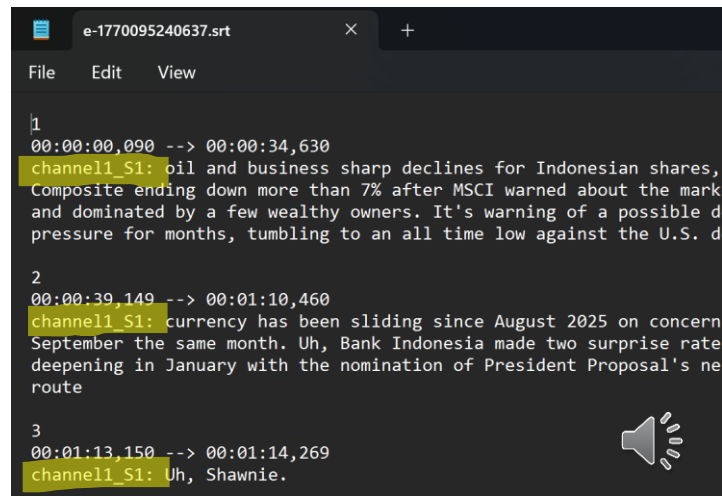


```
e-1770095240637.srt
File Edit View

1
00:00:00,090 --> 00:00:34,630
oil and business sharp declines for Indonesian shares, the benchmark
more than 7% after MSCI warned about the market's instability. U.S.
wealthy owners. It's warning of a possible downgrade if progress
to an all time low against the U.S. dollar.

2
00:00:39,149 --> 00:01:10,460
currency has been sliding since August 2025 on concerns about fiscal
the same month. Uh, Bank Indonesia made two surprise rate cuts in
in January with the nomination of President Proposal's nephew for

3
00:01:13,150 --> 00:01:14,269
Uh, Shawnie.
```



```
e-1770095240637.srt
File Edit View

1
00:00:00,090 --> 00:00:34,630
channel1 S1: oil and business sharp declines for Indonesian shares,
Composite ending down more than 7% after MSCI warned about the mark
and dominated by a few wealthy owners. It's warning of a possible d
pressure for months, tumbling to an all time low against the U.S. d

2
00:00:39,149 --> 00:01:10,460
channel1 S1: currency has been sliding since August 2025 on concern
September the same month. Uh, Bank Indonesia made two surprise rate
deepening in January with the nomination of President Proposal's ne
route

3
00:01:13,150 --> 00:01:14,269
channel1 S1: Uh, Shawnie.
```

Task 2 (Verified transcript edits reflect in downloads)

Magor

Search

This recording

test

Indonesian stocks plunges

Edit Transcript

Transcript

Download

Save

Go to Magor Transcript Studio

00:00:00

Oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Dakarta Composite ending down more than 7% after MSCI warned about the market's investability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.

00:00:39

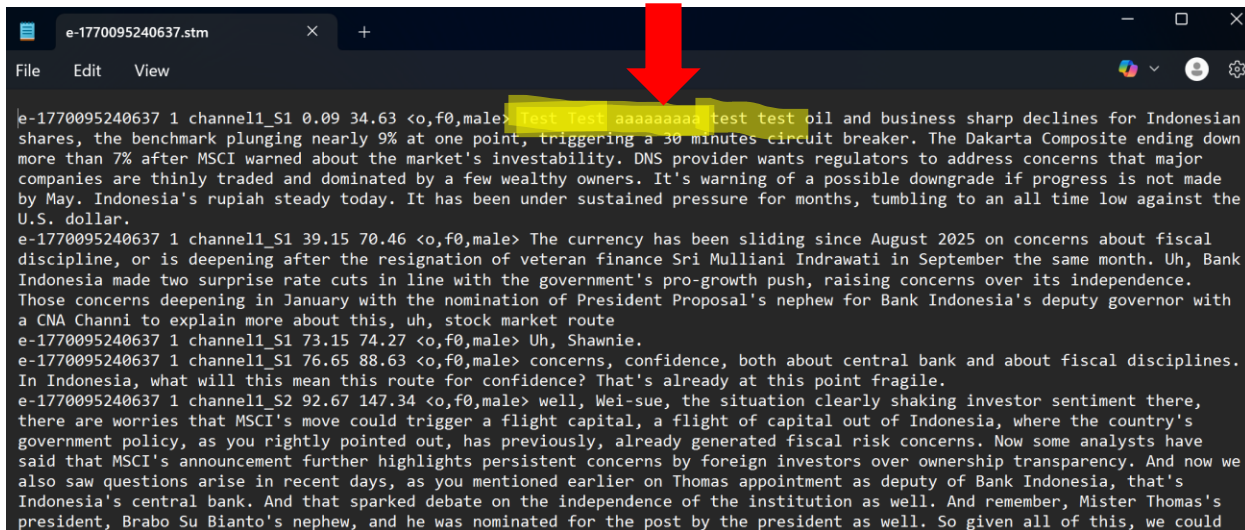
currency has been sliding since August 2025 on concerns about fiscal discipline, or is deepening after the resignation of veteran finance Sri Mulliani Indrawati in September the same month. Uh, Bank Indonesia made two surprise rate cuts in line with the government's pro-growth push, raising concerns over its independence. Those concerns deepening in January with the nomination of President Proposal's nephew for Bank Indonesia's deputy governor with a CNA Channi to explain more about this, uh, stock market route.

00:01:13

Uh, Shawnie.



Task 2 (Verified transcript edits reflect in downloads)



The screenshot shows a web browser window with the address bar displaying 'e-1770095240637.stm'. The browser has a dark theme and standard navigation buttons. The main content area displays a transcript with multiple lines of text. A red arrow points to a specific line in the transcript: 'e-1770095240637 1 channel1_S1 0.09 34.63 <o,f0,male> test test aaaaaaaaaa test test oil and business sharp declines for Indonesian shares, the benchmark plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The Jakarta Composite ending down more than 7% after MSCI warned about the market's instability. DNS provider wants regulators to address concerns that major companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade if progress is not made by May. Indonesia's rupiah steady today. It has been under sustained pressure for months, tumbling to an all time low against the U.S. dollar.'

The transcript continues with several other lines of text, including mentions of 'The currency has been sliding since August 2025 on concerns about fiscal discipline', 'The situation clearly shaking investor sentiment there', and 'So given all of this, we could'.

stm



Task 2 (Verified transcript edits reflect in downloads)

```
e-1770095240637 (2).TextGrid  X  +
File Edit View

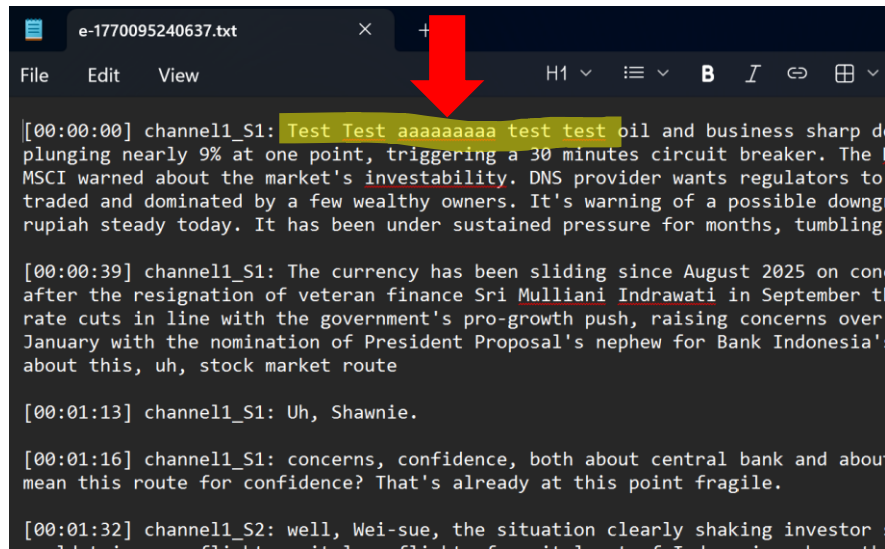
File type = "ooTextFile"
Object class = "TextGrid"

xmin = 0.09
xmax = 229.98
tiers? <exists>
size = 2
item []:
  item [1]:
    class = "IntervalTier"
    name = "channel1_S1"
    xmin = 0.09
    xmax = 229.98
    intervals: size = 11
    intervals [1]:
      xmin = 0.09
      xmax = 34.63
      text = "Test Test aaaaaaaaaa test test oil and business sharp declines for Indonesian shares,
circuit breaker. The Dakarta Composite ending down more than 7% after MSCI warned about the market's inve
companies are thinly traded and dominated by a few wealthy owners. It's warning of a possible downgrade i
been under sustained pressure for months, tumbling to an all time low against the U.S. dollar."
    intervals [2]:
      xmin = 34.63
```

TextGrid



Task 2 (Verified transcript edits reflect in downloads)



The screenshot shows a text editor window titled 'e-1770095240637.txt'. The editor has a dark theme and a menu bar with 'File', 'Edit', and 'View'. A red arrow points to a line of text that is highlighted in yellow. The text in the editor is as follows:

```
[00:00:00] channel1_S1: Test Test aaaaaaaaaa test test oil and business sharp de  
plunging nearly 9% at one point, triggering a 30 minutes circuit breaker. The U  
MSCI warned about the market's investability. DNS provider wants regulators to  
traded and dominated by a few wealthy owners. It's warning of a possible downgr  
rupiah steady today. It has been under sustained pressure for months, tumbling
```

[00:00:39] channel1_S1: The currency has been sliding since August 2025 on cond
after the resignation of veteran finance Sri Mulliani Indrawati in September th
rate cuts in line with the government's pro-growth push, raising concerns over
January with the nomination of President Proposal's nephew for Bank Indonesia's
about this, uh, stock market route

[00:01:13] channel1_S1: Uh, Shawnie.

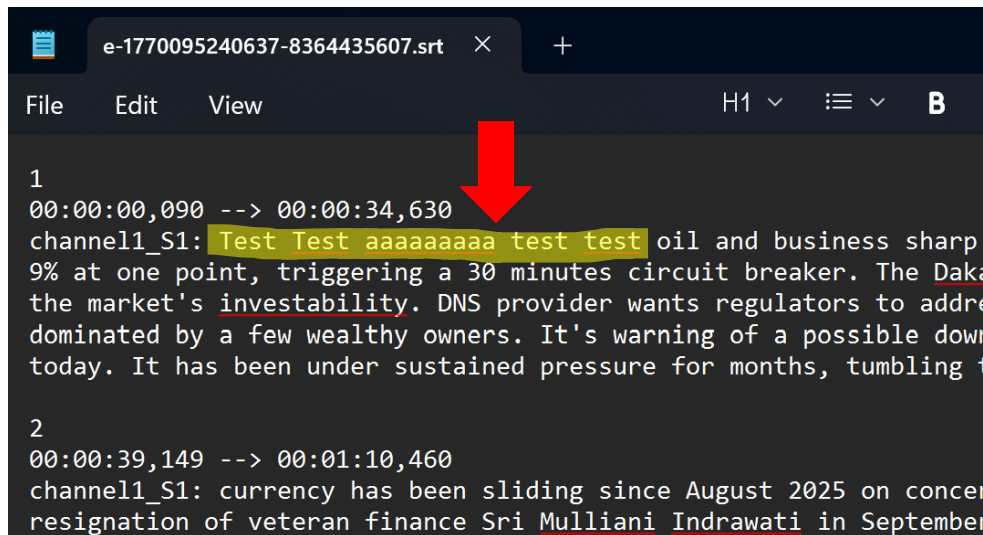
[00:01:16] channel1_S1: concerns, confidence, both about central bank and about
mean this route for confidence? That's already at this point fragile.

[00:01:32] channel1_S2: well, Wei-sue, the situation clearly shaking investor s
could trigger a flight out of Indonesia where th

txt



Task 2 (Verified transcript edits reflect in downloads)



The screenshot shows a text editor window with a dark theme. The title bar indicates the file is 'e-1770095240637-8364435607.srt'. The menu bar includes 'File', 'Edit', and 'View'. The text content is an SRT file with two entries. A red arrow points to the first entry, specifically to the text 'Test Test aaaaaaaaaa test test' which is highlighted in yellow. The text continues with 'oil and business sharp' and '9% at one point, triggering a 30 minutes circuit breaker. The Daka'.

```
1
00:00:00,090 --> 00:00:34,630
channel1_S1: Test Test aaaaaaaaaa test test oil and business sharp
9% at one point, triggering a 30 minutes circuit breaker. The Daka
the market's investability. DNS provider wants regulators to addre
dominated by a few wealthy owners. It's warning of a possible down
today. It has been under sustained pressure for months, tumbling t

2
00:00:39,149 --> 00:01:10,460
channel1_S1: currency has been sliding since August 2025 on concer
resignation of veteran finance Sri Mulliani Indrawati in September
```

srt



Task 3 (Aligned recording list columns)

- Fixed column alignment issues for recording list

Magor		
Title	Created	Recording Type
Indonesian stocks plunges	7 days ago	View Edit Delete
harvard audio	7 days ago	View Edit Delete
speech video test	7 days ago	View Edit Delete
< Previous Page Showing 1 to 3 of 3 result(s) Next Page >		



Magor		
test		
Title	Created	Recording Type
Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	3 days ago	View Edit Delete
Parliament passes motion considering Pri Media	3 days ago	View Edit Delete
Indonesian stocks plunges	11 days ago	View Edit Delete
harvard audio	11 days ago	View Edit Delete
speech video test	11 days ago	View Edit Delete
< Previous Page Showing 1 to 5 of 5 result(s) Next Page >		



Task 3 (Aligned recording list columns)

- ACTUAL FIX: Recording type content display

Magor		
test		

Title	Created	Recording Type
Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	3 days ago	View Edit Delete
Parliament passes motion considering Pri Media	3 days ago	View Edit Delete
Indonesian stocks plunges	11 days ago	View Edit Delete
harvard audio	11 days ago	View Edit Delete
speech video test	11 days ago	View Edit Delete

< Previous Page Showing 1 to 5 of 5 result(s) Next Page >



Magor		
test		

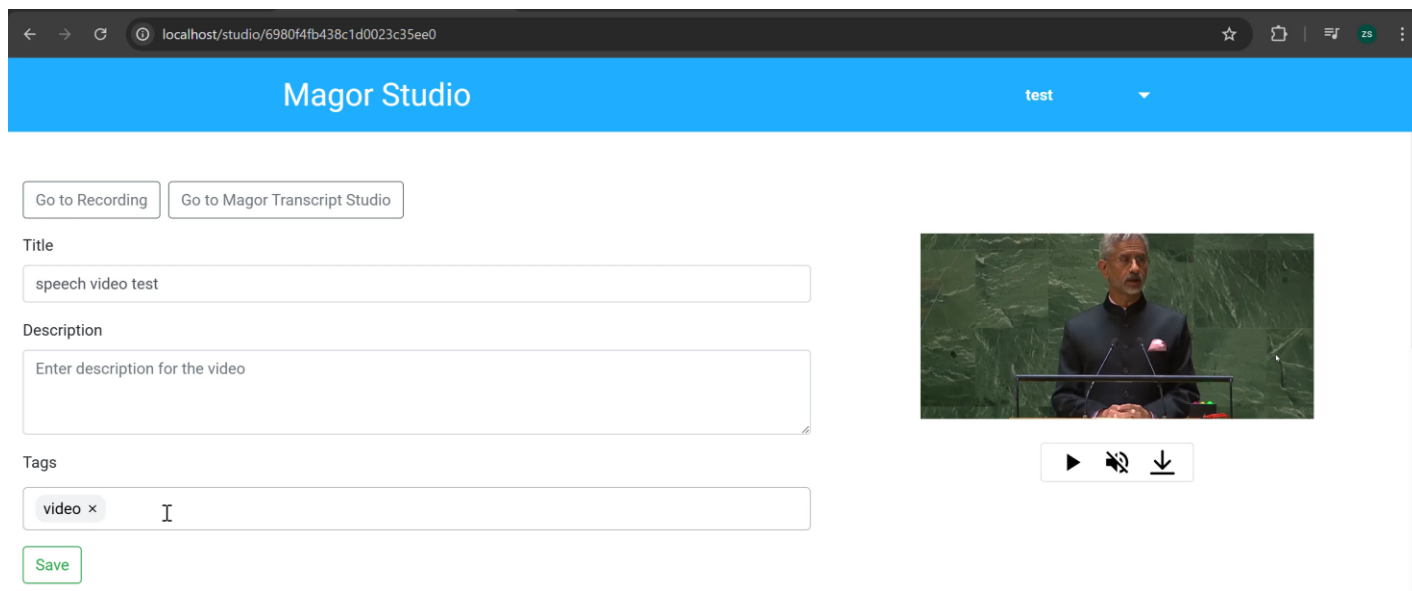
Title	Created	Recording Type	
Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	3 days ago	video/mp4	View Edit Delete
Parliament passes motion considering Pri Media	3 days ago	video/mp4	View Edit Delete
Indonesian stocks plunges	11 days ago	video/mp4	View Edit Delete
harvard audio	11 days ago	audio/wave	View Edit Delete
speech video test	11 days ago	video/mp4	View Edit Delete

< Previous Page Showing 1 to 5 of 5 result(s) Next Page >



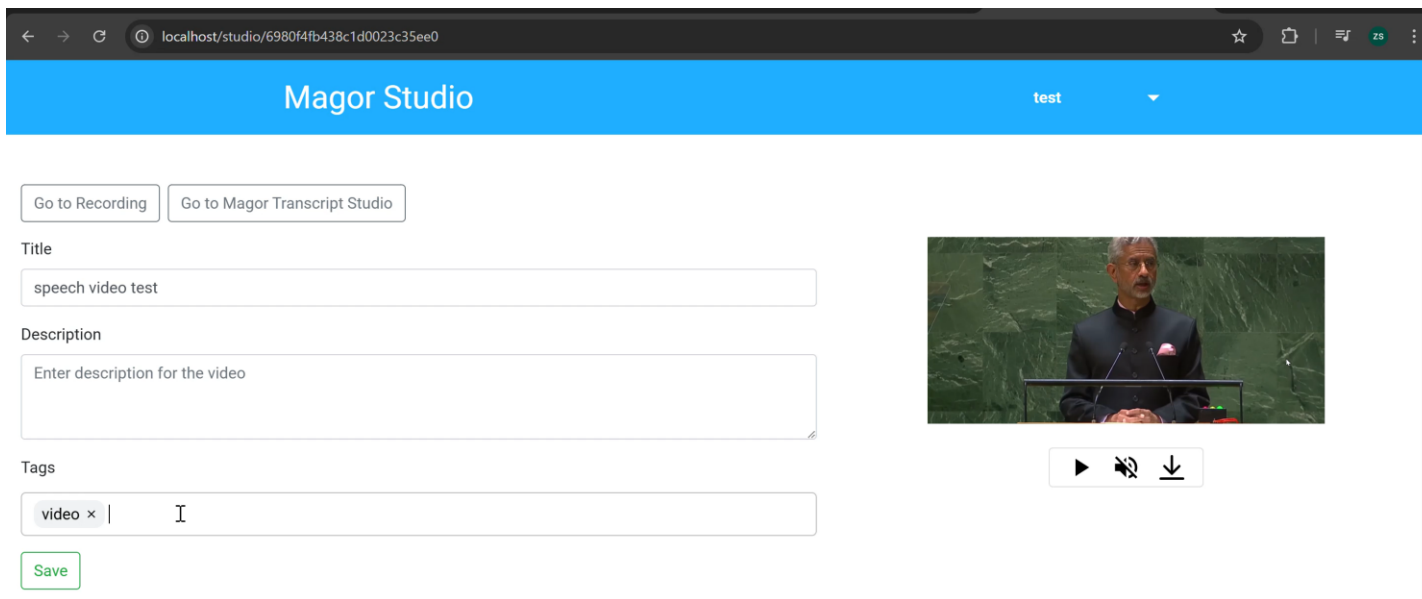
Task 4 (Fixed unintended tag removal on backspace)

Bug Demo:



Task 4 (Fixed unintended tag removal on backspace)

Fixed Tagging Demo:



The screenshot shows a web browser window with the address bar displaying `localhost/studio/6980f4fb438c1d0023c35ee0`. The page title is "Magor Studio" and the user is logged in as "test".

At the top, there are two buttons: "Go to Recording" and "Go to Magor Transcript Studio".

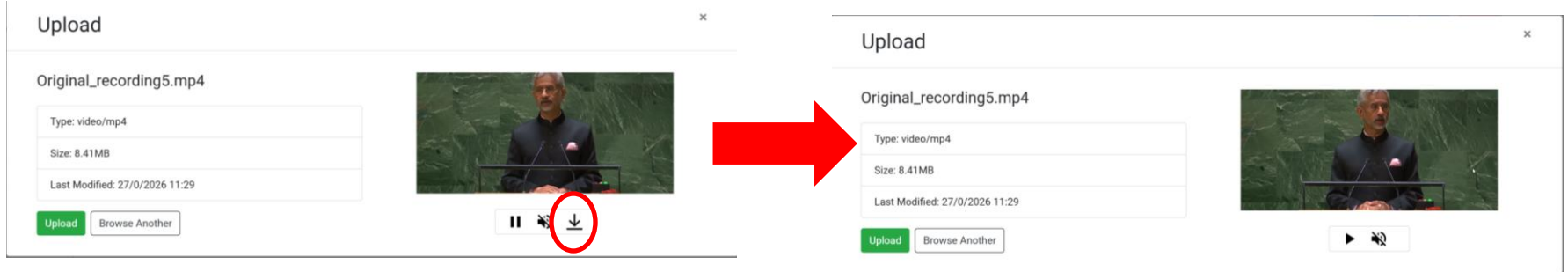
The main form contains the following fields:

- Title:** A text input field containing "speech video test".
- Description:** A text area with the placeholder text "Enter description for the video".
- Tags:** A text input field containing "video x |" followed by a cursor. Below this field is a green "Save" button.

On the right side of the form, there is a video player showing a man in a dark suit speaking at a podium. Below the video player are three icons: a play button, a mute button, and a download button.

Task 5 (Removed download button when uploading media)

- Unintended download button when uploading media



Plans for Next Week

- Fix Textgrid conversion
 - Replace xmin=0.33 to 0
- Add multi-select and “Download All” options for transcript downloads
- Fix video playback subtitle overflow and incorrect display issues
- Implement transcript download in Magor Transcript Studio page

